Grade 11 – Term 1
Choosing a Statistical Model: C10 Practice

1 Your Assignment

Choose **1** of the 3 options below and write an **individual academic PDF report analyzing 1 real-life situation** using your selected statistical model. Independently identify the situation and find **1 credible, publicly accessible dataset containing actual observations**. Do not use invented or simulated datasets or copy completed practice problems.

In the report, justify your selected model using the data and context, not simply a convenient formula. Apply methods studied in C1–C9 and show mathematical procedures even when using software.

2 Choose 1 Option

1. **Linear probability density model.** Investigate a continuous variable that can reasonably be modeled by a linear probability density function over a stated interval. Specify its density, parameters, and support; verify nonnegativity and total area 1.
2. **Exponential model.** Investigate waiting times or intervals between events that can reasonably be modeled by an exponential distribution. Specify or estimate the rate parameter and justify a constant event rate and independent waiting times.
3. **Normally distributed measurements.** Investigate measurements that can reasonably be modeled by a normal distribution. Use the normal distribution or Student's t for inference about the population mean, as appropriate. State an inferential question and interpret your analysis in context. Justify the sampling assumptions and the plausibility of a normal population. Student's t is used for inference about a population mean when the population standard deviation is unknown and estimated from the sample, with $\nu = n - 1$ degrees of freedom. Under appropriate assumptions, it describes the standardized sample-mean statistic; it is **not the distribution of the original observations merely because the sample is small**.

3 During the Exam

Select **1** option and write a very brief description (1–2 sentences) of real-life data that could plausibly follow your selected distribution. This C10 response is worth **6 points**. No datasets, sources, calculations, parameter estimates, or detailed model justification are required during the exam.

4 Requirements for the Online PDF Report

Your PDF report will focus on the **1 option selected during the exam** and must be submitted virtually through **Open**. This online submission is separate from the brief response written during the exam. Complete the following 8 requirements for your **1 selected situation** in the report:

1. **Describe the context.** Explain the real-life situation, define the random variable and its units, identify the population and relevant time period, and state the objective of the analysis.

2. **Find real data.** Provide a working, clickable hyperlink to the original database or dataset. Identify the source, sample size, relevant variables, and how observations were collected or selected. Explain any filtering or missing-value treatment. Cite the source where used.
3. **Organize the data.** Present the observations or an appropriate frequency table with clear headings and units. For a large dataset, use a frequency table and explain how it was constructed.
4. **Create a histogram.** Construct a readable histogram from the dataset with meaningful class intervals, frequencies, a descriptive title, and labeled axes and units. Explain your interval choices. If overlaying a theoretical curve, use compatible scales: a density histogram or a curve rescaled to frequencies.
5. **Select and justify the model.** Specify the distribution, support, and parameters, distinguishing known values, estimates, and assumptions. Justify the choice using the context, data characteristics, and histogram. Discuss assumptions and limitations. A histogram provides empirical evidence but **does not, by itself, establish that a theoretical distribution is appropriate**. Explain any important mismatch and qualify your conclusions.
6. **Calculate sample statistics.** Calculate the sample mean, sample variance, and sample standard deviation using

$$\bar{x} = \frac{1}{n}\sum_{i=1}^{n} x_i, \qquad s^2 = \frac{1}{n-1}\sum_{i=1}^{n}(x_i - \bar{x})^2, \qquad s = \sqrt{s^2}.$$

   Show substitutions and intermediate steps; software results alone are insufficient. State units, including squared units for variance. Prefer original observations for calculations; if using grouped data, show the frequency-based procedure and label midpoint-based results as approximations.
7. **Calculate theoretical statistics.** Whenever parameters can reasonably be specified or estimated, determine the model's theoretical mean, variance, and standard deviation using the formulas or methods studied. Show parameters, formulas, substitutions, and procedures. Clearly distinguish model values from sample statistics and label values based on estimated parameters as estimates. If you select normally distributed measurements, compare observations with the underlying normal measurement model; keep Student's t inference separate. If a parameter cannot reasonably be determined, explain the missing information and how it limits the comparison.
8. **Compare and interpret.** Compare the histogram and sample statistics with the theoretical model. Explain agreements and differences, considering sampling variation, parameter estimation, and assumptions that may fail. If fitting makes a model statistic equal to a sample statistic, acknowledge that this equality is not independent evidence of fit. Discuss whether the evidence supports the model and interpret the findings and limitations in context.

5 Required Report Structure

Write a connected academic report using the following structure. Integrate data, graphs, and calculations into the discussion rather than submitting disconnected answers.

Title and Author Give a descriptive title and your full name.

Introduction Explain the report's purpose and introduce your selected situation and its objective.

Methodology Describe the dataset, source, variables, sample size, data preparation, model selection, assumptions, parameter estimates, and statistical procedures.

Results and Discussion Identify your selected model clearly and include the data or frequency table, histogram, calculations, justification, comparisons, interpretations, and limitations required above.

Conclusion Summarize the findings and discuss the usefulness and limitations of your selected model.

References Give complete citations and clickable hyperlinks to your original dataset: author or organization, dataset title, date or version where available, and access date. Cite any other sources and tools used, and connect in-text citations to the reference list.

6 Before You Submit

Submit 1 **readable, professionally organized PDF report written individually** virtually through **Open**. Check that the report focuses on your **1 selected option**, uses real data, justifies the model choice, and shows mathematical procedures. Use clear headings, complete paragraphs, and consistent notation. Open the final PDF to verify that all graphs, tables, and equations are complete and readable. Test the original dataset hyperlink to confirm that it works.